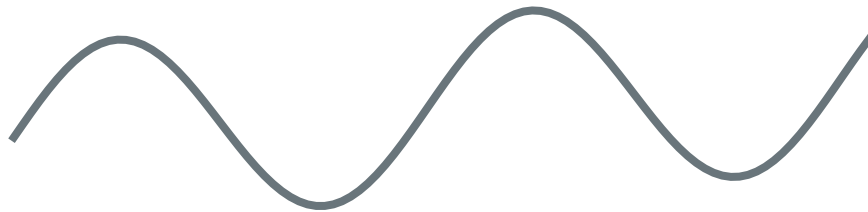


NEURAL NETS LEARN CURVES. MINE LOST TO A FLAT LINE.

I used quarterly bank reports to predict how deposits would change next quarter.



NEURAL NET

3.627



0% FOR EVERY BANK

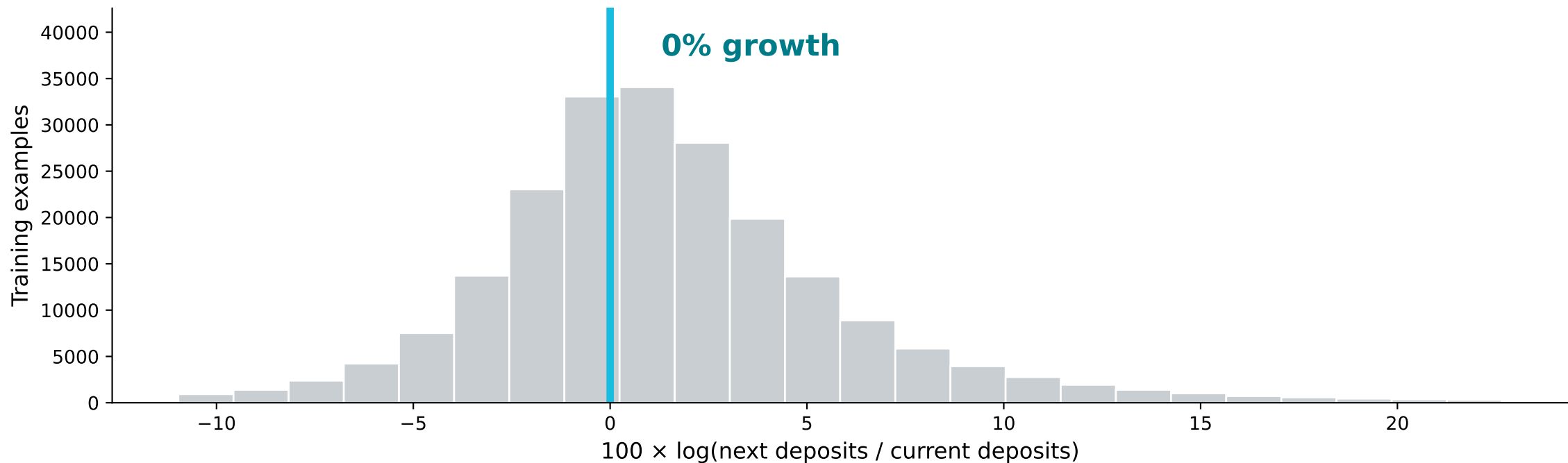
3.601

Lines illustrate the two ideas; they are not plotted predictions.

Average absolute forecast miss (MAE), percentage points. Lower is better. Original four-model comparison. 214,425 training examples; 13,532 evaluation examples. The small gap does not establish a future winner.

WHY WAS A FLAT LINE SO HARD TO BEAT?

A LOT OF DEPOSIT CHANGES SIT NEAR ZERO.



BORING CAN WORK.

Central 98% of training log-growth targets. Full tails stayed in the experiment.
Training concentration motivates the baseline; the later evaluation establishes its score.

**ZERO WON THE AVERAGE MISS.
THE NET WON WHEN BIG MISSES HURT MORE.**

AVERAGE MISS ↓

0% 3.601

Neural net 3.627

BIG MISSES HURT MORE ↓

Neural net 6.429

0% 6.718

“WINNER” DEPENDS ON WHAT HURTS.

MAE on the left; RMSE on the right. Both measure ordinary-growth forecast errors in percentage points.
Same evaluation rows, original four models. Lower RMSE does not mean every large miss improved.

THE FLAT LINE COULDN'T TELL ME WHERE TO LOOK.

BANK A 0% BANK B 0% BANK C 0% BANK D 0%

EVERYBODY IS TIED.

4,536 BANKS. 454 REVIEW SLOTS.

RANDOM SELECTION

45.4

expected

RIDGE

112

found

NEURAL NET

109

found

NOW I HAVE A LIST.

March → June 2024. Hypothetical 10% review capacity. Found = selected bank later in the lowest-growth 10%. That group can include positive growth. A human still investigates the cause; no savings were measured.

I THREW OUT 4 INPUTS. KEPT ONLY DEPOSIT SIZE.

Deposit size = customer deposit dollars. Both comparisons use Ridge.

1 INPUT

18.8

+3.1

per 100 selected

5 INPUTS

21.9

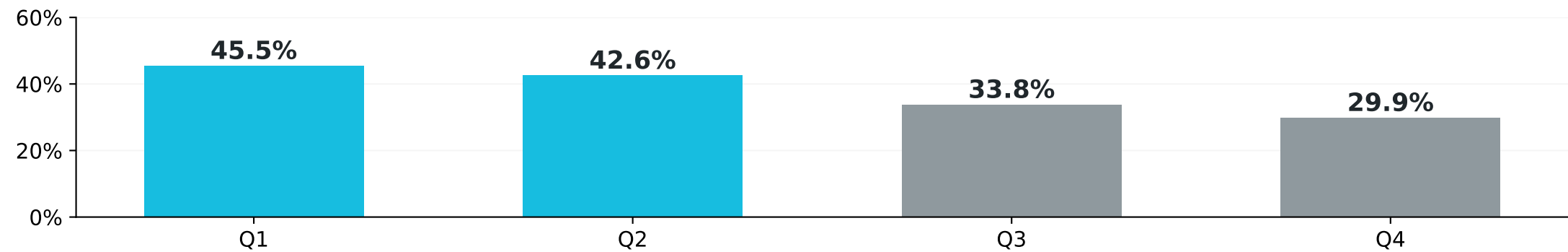
Random selection $\approx$ 10

THAT'S IT?

Lowest-growth banks found per 100 selected; equal-quarter averages over three reused 2024 quarters. Size-only Ridge was a later challenger. This is not a causal decomposition; the gap needs a fresh test.

A CALENDAR RULE BEAT THE FLAT LINE.

How often did deposits fall next quarter? Training years:



SEASONAL RULE

3.500

FLAT LINE

3.601

SERIOUSLY?
THE NEXT RESULT I CARE ABOUT
IS THE ONE I DON'T KNOW YET.

Freeze these rules. Test the next unseen quarter.

Seasonal median was developed after examining 2024. It predicts each calendar quarter's training median. Historical MAE in percentage points; lower is better. Analysis notebook built with wm-notecards.